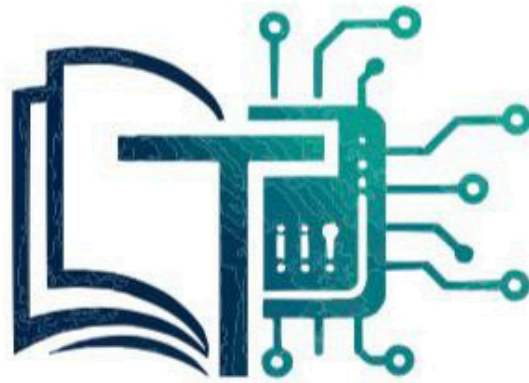




Tech & Teach

Digital Chip Design Program

RTL Design

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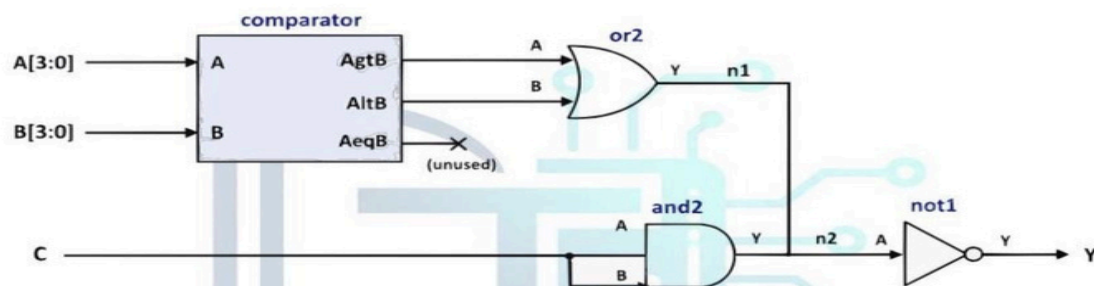
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7/07/2026

VERSION	ACTION	DATE	AUTHOR
1.0	Initial draft	05/07/2026	Abdulmalek Aldossery

Task #1.1

Using SystemVerilog, implement the following design by connected the blocks (USE the design source below)

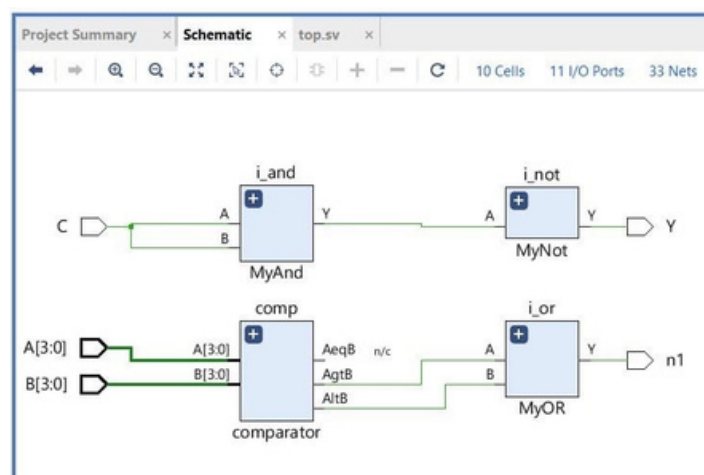
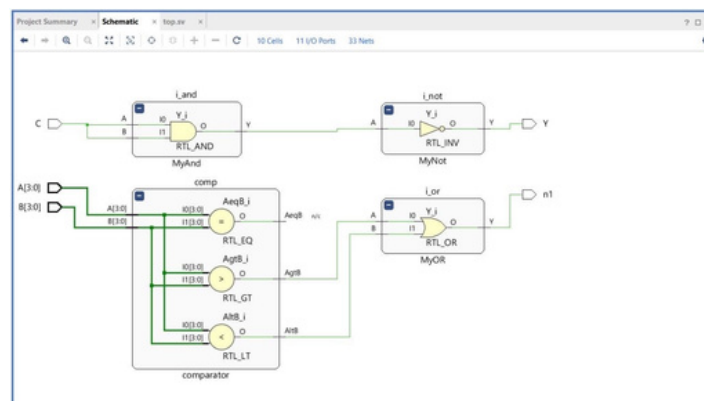


comparator.sv:

```

1 module comparator(AeqB, AltB, AgtB, a, b);
2
3 input [3:0] a,b;
4 output AeqB, AltB, AgtB;
5
6 assign AeqB = (A == B)? 1 : 0;
7 assign AltB = (A < B)? 1 : 0;
8 assign AgtB = (A > B)? 1 : 0;
9 endmodule

```



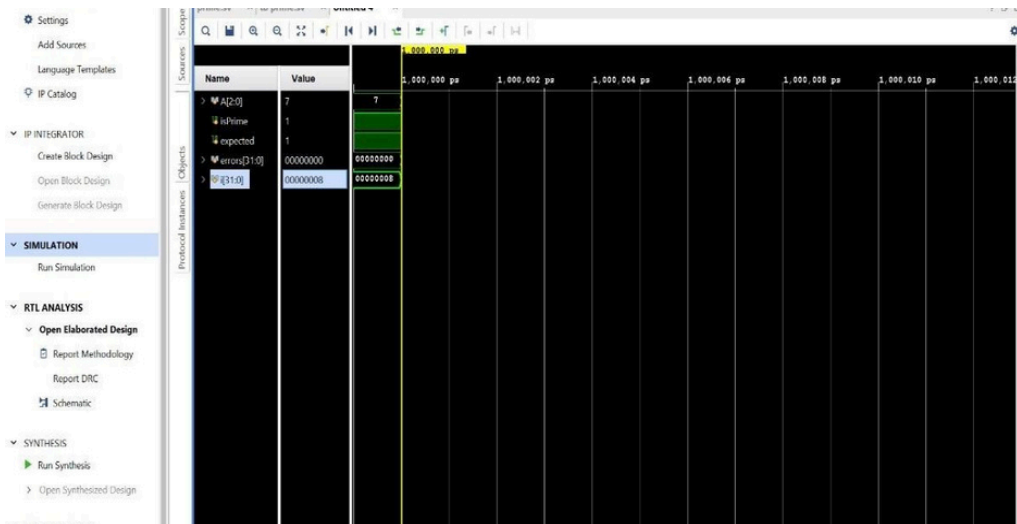
Task #1.2

Given the following SystemVerilog design for prime Circuit that check the 3-bit binary number is prime or not (Prime numbers are those number that can be divided by only 1 or itself like 2,3,5,7). You are asked to write the test bench to verify the design (USE the design source below):

A2	A1	A0	isPrime
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

prime.sv:

```
1 module prime(  
2  
3 input logic [2:0] A,  
4 output logic isPrime  
5  
6 );  
7  
8 assign isPrime = (A[2])? A[0]: A[1];  
9  
10 endmodule
```

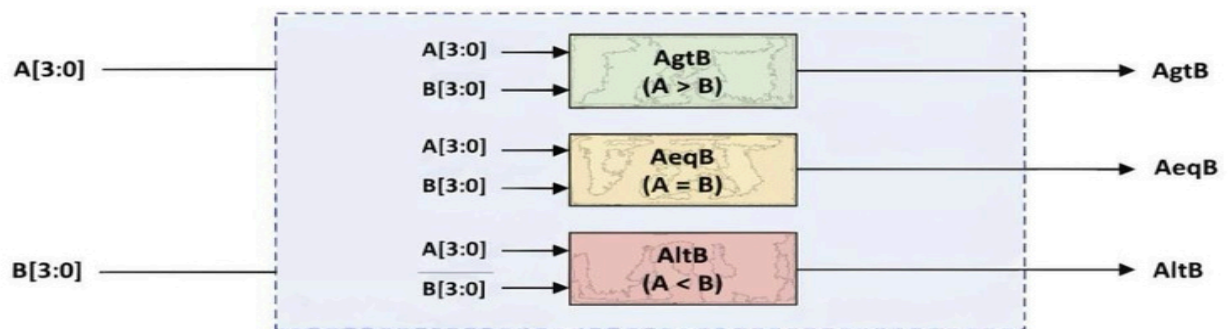


```
1  
2  
3 module tb_prime;  
4  
5     logic [2:0] A;  
6     logic      isPrime;  
7     logic      expected;  
8  
9     // Instantiate DUT  
10    prime DUT (  
11        .A      (A),  
12        .isPrime (isPrime)  
13    );  
14  
15    // Reference model based on the truth table in the task  
16    function automatic logic is_prime_ref(input logic [2:0] val);  
17        case (val)  
18            3'b000: is_prime_ref = 1'b0;  
19            3'b001: is_prime_ref = 1'b0;  
20            3'b010: is_prime_ref = 1'b1;  
21            3'b011: is_prime_ref = 1'b1;  
22            3'b100: is_prime_ref = 1'b0;  
23            3'b101: is_prime_ref = 1'b1;  
24            3'b110: is_prime_ref = 1'b0;  
25            3'b111: is_prime_ref = 1'b1;  
26        endcase  
27    endfunction
```

Task #1.3

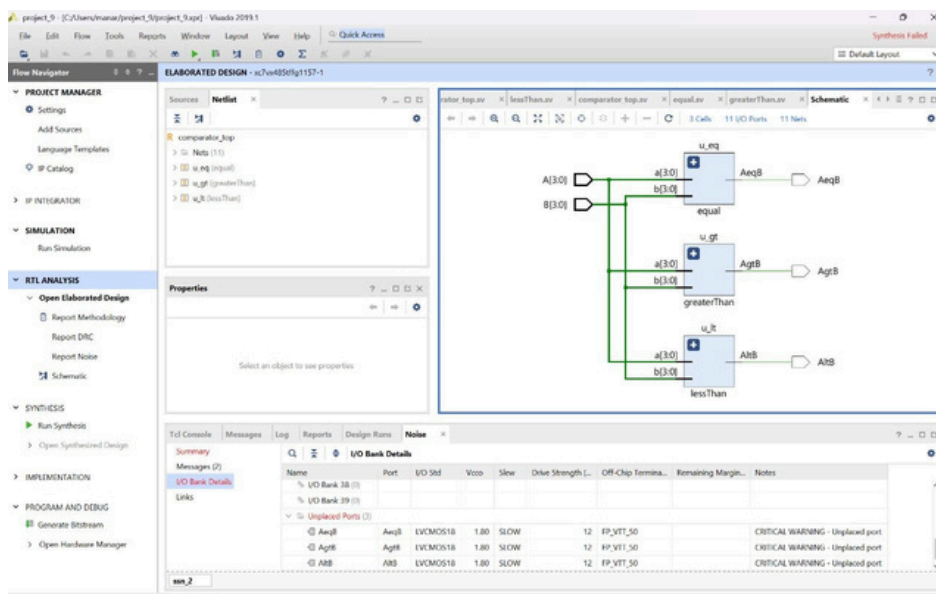
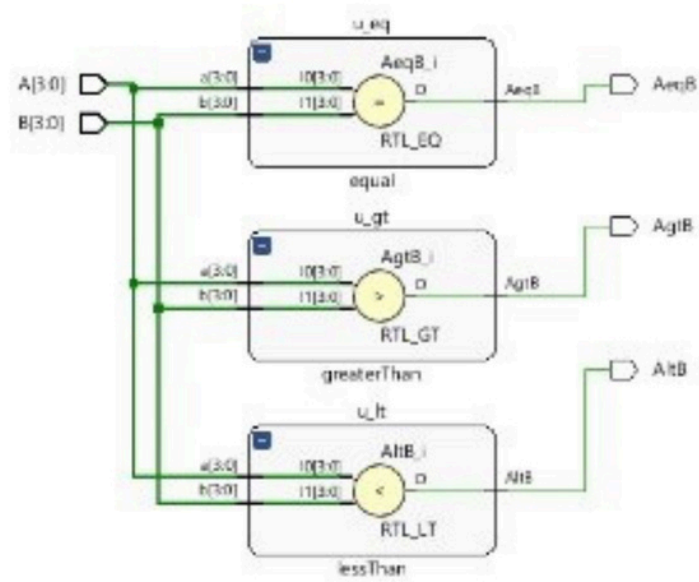
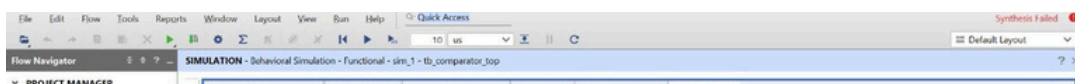
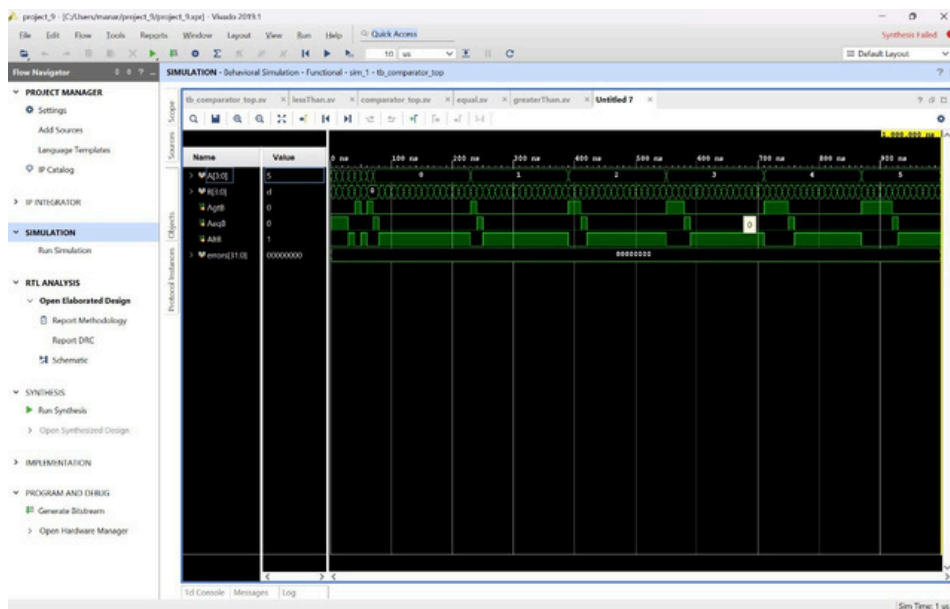
Using SystemVerilog, implement the following design and (USE the design sources below) then write testbench to verify the comparator

A, B: unsigned binary numbers
If A==B, then AeqB=1, AltB=0, AgtB=0
If A<B, then AeqB=0, AltB=1, AgtB=0
If A>B, then AeqB=0, AltB=0, AgtB=1



lessThan.sv:

```
1
2 module lessThan(AltB, a, b);
3
4 input [3:0] a,b;
5 output AltB;
6
7 assign AltB = (A < B)? 1 : 0;
8 endmodule
9
```



section	alia	manar	aryam	elaf	lana
report	✓	✓	✓	✓	✓
task-1-1	✓	✓	✓	✓	✓
task-1-2	✓	✓	✓	✓	✓
task-1-3	✓	✓	✓	✓	✓